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Creating Electronic Books with Code and Artificial Intelligence Assistants

Course for the Faculties of Sciences and Chemical Sciences

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Welcome to **Creating Electronic Books with Code and Artificial Intelligence Assistants**.

What is this?

This is the course material and a template designed for the teaching staff of the **Faculties of Sciences and Chemical Sciences at USAL** to create interactive teaching books easily. The main idea of this template is for it to be broad enough to cover many use cases and for each student to adapt it to their own course, while also being sufficiently equipped to be used easily with the help of AI assistants (such as GitHub Copilot, Gemini, Claude, Codex, etc.) and a code editor such as VS Code.

Content

In this book you will find:

- *Tutorials* to learn how to use the template
- *Examples by Degree* to see real cases
- Information on *how to cite* and *licenses*

PDF version

You can also download the printable version of the book:

- Download PDF in English
- Descargar PDF en español

Note

This project is designed to be used with **VS Code** and **AI** assistants.

CENTRAL LIMIT THEOREM

The Central Limit Theorem (CLT) states that the distribution of sample means approximates a normal distribution as the sample size gets larger, regardless of the population's actual distribution shape, provided the variance is finite.

1.1 Mathematical Expression

Let $X_1, X_2, \dots, X_n$ be a random sample of size n drawn from a population with mean μ and variance σ^2 . Then, the random variable:

$$Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \quad (1.1)$$

converges in distribution to a standard normal $N(0, 1)$ as $n \rightarrow \infty$.

Important

The sample size n is generally considered large enough to apply the CLT when $n \geq 30$.

BIBLIOGRAPHY